

PUSHKAR KATHAYAT

Senior Full-Stack Engineer | Python, TypeScript, AI Systems

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PROFESSIONAL SUMMARY

Full-stack engineer with **8+ years of experience** shipping production systems end to end: Python (FastAPI, Django) backends, TypeScript (React, Next.js) frontends, PostgreSQL schemas, queues, and the CI/CD and cloud infrastructure they run on. Currently building multi-agent LLM products in production, with RAG, evaluation and review gates, deployed on AWS ECS and Linux servers behind test-gated pipelines. Former CTO of an AI SaaS at 18,000 users. Merged contributor to a leading open-source AI agent platform. Comfortable owning a system from design through release and on-call.

SELECTED HIGHLIGHTS

- Built the Python backend for a production **multi-agent simulation platform orchestrating 18 LLM agents** (Claude, GPT-5, Perplexity), deployed to **AWS ECS Fargate via CodePipeline** with separate staging and production stacks
- Wrote **1,200+ pytest tests** around agent orchestration, evaluation and calibration logic, so prompt and model changes are caught before release
- Run **13 production apps across 8 domains** on Linux (nginx, pm2, PostgreSQL, Redis), shipped through GitHub Actions pipelines that block deploys on failing tests and migration drift
- Tuned a core PostgreSQL read path from **1.2s to 180ms** and cut API response times from **2s to 300ms** with Redis caching

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL
Backend	FastAPI, Django, DRF, Node.js, Pydantic, Celery, arq, BullMQ, REST, tRPC, WebSockets, OAuth 2.0, webhooks
Frontend	React, Next.js (App Router), TypeScript, React Query, Tailwind, shadcn/ui
Databases	PostgreSQL (schema design, migrations, indexing), MongoDB / DocumentDB, Redis, Kysely, Drizzle
Cloud & DevOps	AWS (ECS Fargate, ECR, CodePipeline, CodeBuild, Amplify, Lambda, S3, SSM, ALB), Cloudflare (Workers, Pages), Docker, docker-compose, GitHub Actions, nginx, pm2, Linux server administration
AI & LLMs	Multi-agent orchestration, RAG (knowledge bases, MMR retrieval), Claude / OpenAI / Gemini / Perplexity APIs, tool calling, MCP servers, LLM evaluation harnesses
Testing	pytest, Vitest, integration and smoke tests in CI, regression-first bug fixing
Practices	System design, production deployments, staging/prod separation, infra and secrets security audits, AI-assisted development with review gates

PROFESSIONAL EXPERIENCE

Senior Full-Stack & AI Engineer AI Agents & LLM Products Remote	Mar 2026 - Present
- Built the Python FastAPI backend for a client-facing multi-agent simulation platform, orchestrating 18 LLM agents across Claude, GPT-5 and Perplexity with coherence and review gates between stages, plus RAG over client documents (PDF, DOCX, web) with MMR retrieval - Shipped it on AWS: Docker images built in CodeBuild, pushed to ECR and rolled out to ECS Fargate behind an ALB through CodePipeline, with separate staging and production stacks - Built the evaluation, calibration and prompt-variant layer and covered it with 1,200+ pytest tests; Next.js / TypeScript frontend auto-deployed via AWS Amplify - Built a client deliverable portal in Next.js and TypeScript on PostgreSQL (94 migrations) with Redis/BullMQ workers and an MCP server exposing 35 tools to AI clients - Built an LLM visibility audit product in Python FastAPI with arq/Redis workers, scoring brands across Gemini, GPT-4.1, Claude and Perplexity, covered by ~470 pytest tests - Set up GitHub Actions CI/CD gating every deploy on a Postgres-backed test run, migration drift check, build and smoke tests; run 13 production apps across 8 domains (nginx, pm2, PostgreSQL, Redis) and wrote infra and secrets security audits	

Backend / Full-Stack Engineer | Real Jobs From Anywhere | Remote (Contract) Jul 2025 - Nov 2025

- Architected a job aggregation platform unifying **5,000+ records** from heterogeneous sources into one normalized schema
- Built a **7-queue BullMQ system processing 10,000+ jobs daily** with retries and dead-letter handling; Python Scrapy ingestion on AWS Lambda
- Designed the multi-tenant PostgreSQL schema; composite indexes took a core query from **1.2s to 180ms**

Full-Stack Developer | YouTube Transcript Generator | Remote (Freelance) Jan 2024 - Nov 2024

- Built a transcription pipeline on Python FastAPI with background jobs and a Next.js frontend; Redis caching cut upstream API costs **60%** and response times from **2s to 300ms**

Chief Technology Officer | Semantic Pen | Remote Jan 2023 - Dec 2023

- Owned architecture for an AI content SaaS serving **18,000 users** and generating 10,000+ articles; **led 3 engineers**
- Built the RAG pipeline with vector search, scaled the backend on AWS (Lambda, ECS, S3, SES), and integrated Paddle billing with webhooks and usage metering

Backend / Automation Engineer | Hero Asian Kitchen, Vintage Kitchen | Full-time Jan 2021 - Mar 2023

- Built unattended data pipelines in Python (Scrapy, FastAPI, AWS Lambda) and integrated early LLM APIs into production workflows

Founder | Degions | India Nov 2018 - Dec 2020

- Built and sold developer tooling to thousands of users, owning product, deployment, billing and support

SELECTED PROJECTS & OPEN SOURCE

Open-source AI agent platform (389k+ GitHub stars)

Merged fixes to its Discord and Slack integrations, including forum tag support carried from the agent tool through the channel action to the Discord API, with a regression test proven to fail on the previous code.

Tech: TypeScript, Node.js, Vitest

LLM Evaluation Harness

Benchmarks Claude and GPT models on accuracy, latency and cost per correct answer, turning model selection into a measured tradeoff.

Tech: Python, LLM APIs, evaluation and cost instrumentation

EDUCATION

B.Tech, Computer Science Engineering | THDC-IHET

2016